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Machine Learning

Loss Functions

Loss Function

- The loss function helps determine how effectively your algorithm model the featured dataset.
- Cost functions are vital in machine learning, measuring the disparity between predicted and actual outcomes.
- These functions shape model performance and guide optimization techniques like gradient descent, leading to better predictions.
- The **hyperparameters** are adjusted to minimize the average loss — we find the weights w and biases b , that minimize the value of J (average loss).

$$J(w^T, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)})$$

Types of Loss Function

In supervised learning, there are two main types of loss functions — these correlate to the 2 major types of neural networks: regression and classification.

Regression Loss Functions — given an input value, the model predicts a corresponding output value (rather than pre-selected labels)

- Mean Squared Error
- Mean Absolute Error
- Huber Loss

Classification Loss Functions — given an input, the network produces a vector of probabilities of the input belonging to various pre-set categories — can then select the category with the highest probability of belonging.

- Binary Cross-Entropy
- Categorical Cross-Entropy

Mean Squared Error / L2 Loss

The difference is squared, which means it does not matter whether the predicted value is above or below the target value; however, values with a large error are penalized more.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2$$

Advantage

- **Easy Interpretation:** The MSE is straightforward to understand.
- **Always Differential:** Due to the squaring, it is always differentiable.
- **Single Local Minimum:** It has only one local minimum.

Disadvantage

- **Error Unit in Squared Form:** The error is measured in squared units, which might not be intuitively interpretable.
- **Not Robust to Outliers:** MSE is sensitive to outliers.

Mean Absolute Error / L1 Loss

MSE is highly sensitive to outliers, which can dramatically affect the loss because the distance is squared. MAE is used in cases when the training data has a large number of outliers to mitigate this.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y^{(i)} - \hat{y}^{(i)}|$$

Advantage

- **Intuitive and Easy:** MAE is easy to grasp.
- **Error Unit Matches Output Column:** The error unit is the same as the output column.
- **Robust to Outliers:** MAE is less affected by outliers.

Disadvantage

- **Graph Not Differential:** The MAE graph is not differentiable, so gradient descent cannot be applied directly.

Huber Loss

If the absolute difference between the actual and predicted value is less than or equal to a threshold value, δ , then MSE is applied. Otherwise — if the error is sufficiently large — MAE is applied.

$$Huber = \frac{1}{n} \sum_{i=1}^n \frac{1}{2} (y_i - \hat{y}_i)^2 \quad |y_i - \hat{y}_i| \leq \delta$$

$$Huber = \frac{1}{n} \sum_{i=1}^n \delta \left(|y_i - \hat{y}_i| - \frac{1}{2} \delta \right) \quad |y_i - \hat{y}_i| > \delta$$

- **n**: The number of data points.
- **y**: The actual value (true value) of the data point.
- **$\hat{y}$** : The predicted value returned by the model.
- **δ** : Defines the point where the Huber loss transitions from quadratic to linear.

Binary Cross Entropy/Log Loss

It is used in binary classification problems with two classes.

Binary cross entropy compares each of the predicted probabilities to the actual class output which can be either 0 or 1. It then calculates the score that penalizes the probabilities based on the distance from the expected value. That means how far from the actual value.

$$CE\ Loss = \frac{1}{n} \sum_{i=1}^N - (y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i))$$

True label: 1

Predicted probability: 0.9

$$L = -\log(0.9) = 0.105$$

True label: 1

Predicted probability: 0.1

$$L = -\log(0.1) = 2.302$$

Cross Entropy

Cross entropy is used for Multiclass classification.

Binary cross-entropy is a special case of categorical cross-entropy, where $M = 2$ (the number of categories is 2).

$$L = - \sum y_i \log(\hat{y}_i)$$

True: Class = 1

Predicted probabilities:

- [0.7, 0.2, 0.1]

Loss:

$$L = - \log(0.2) = 1.609$$

Custom Loss Functions

- Mean Squared Logarithmic Error
- Mean Absolute Percentage Error
- Binary Focal Cross-entropy Loss
- Sparse Categorical Cross-entropy Loss
- Kullback-Leibler (KL) Divergence Loss
- Hinge Loss
- Categorical Hinge Loss
- Cosine Similarity

Thank you